

Sana Pandey

Scalable interpretability and evaluation for AI beneficial to the public.

MIT CSAIL · Anthropic Research Fellow · NSF Graduate Research Fellow · UC Berkeley BAIR

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Education

- Massachusetts Institute of Technology** — Graduate Student, EECS. Dual SM in Technology Policy. 2025–2027
- Fully supported by the NSF Graduate Research Fellowship. CSAIL Language and Intelligence Group (with Prof. Jacob Andreas).
 - Coursework:** Reinforcement Learning; Modeling with Machine Learning; Algorithmic and Human Decision-Making; Ethical Machine Learning and Human Deployments; Economic Analysis for Business Decisions.
- University of California, Berkeley** — Bachelor's in Data Science and Cognitive Science 2024
- 3.88/4.0 GPA; minor in Mandarin Chinese; Captain, Women's Epee Fencing Team.
 - Coursework:** Artificial Intelligence; Modeling, Learning, and Decision Making; Data Structures; Structure and Interpretation of Computer Programs; Linear Algebra; Discrete Mathematics and Probability; Designing Algorithmic Media (audited graduate course).
- Stanford University** — China Scholars Program 2020
- Thesis: *Rice Rabbit: Analyzing the Evolution of China's Feminist Movement with the Rise of Censorship and Social Media*. Studied civic participation outcomes following algorithmic censorship interventions.

Experience

- Anthropic** — MIT x Anthropic AI for Social Impact Research Fellow Summer 2026
- Building** skills to support reasoning in expert, high-impact domains. Deployed financial advising tool to over 800,000 people. Co-sponsored by MIT (Prof. Sendhil Mullainathan) and Next Ladder Ventures.
 - Selected** for the inaugural cohort (8 graduate students) to translate research agendas into applied tools for economic mobility with the Beneficial Deployments & Applied AI team.
- MIT CSAIL (LINGO)** — Graduate Research Assistant (Prof. Jacob Andreas) 2025–Present
- Developing** mechanistic-interpretability methods (TransformerLens, activation patching, sparse autoencoders) to predict how data features drive model generalization failures and behaviors.
 - Fine-tuning** open-weight LLMs (PEFT/LoRA) on multi-GPU Slurm clusters, served with vLLM for large-scale evaluation; constructing optimization metrics that encode human values in deployment settings.
- Berkeley AI Research, Center for Human-Compatible AI** — Research Engineer, Applied AI 2024–2025
- Built** white-box evaluations of closed-source LLMs to advance explainability and value alignment; designed evaluation frameworks grounded in mechanism design, game theory, and social choice theory to measure models in collective and democratic decision-making contexts.
- Stealth Startup** — Technical Lead, AI Citizen-Support Systems Fall 2024
- Designed** evaluation metrics for a reinforcement-learning citizen-support assistant improving the interface between residents and city services, piloted with Boston and San Jose as part of a two-person founding team; advised responsible AI integration into municipal services.
- Berkeley AI Research, Center for Human-Compatible AI** — Research Intern 2022–2024
- StrongREJECT:** evaluated eight jailbreak strategies and defined categories of high-risk model output to benchmark refusal robustness.
 - Prosocial Ranking Challenge:** built infrastructure to test prosocial recommenders on live social platforms, cutting modeled polarization by up to two years; implemented reference NLP, statistical, and LLM rankers for participants (featured on CBS Evening News).
 - Non-Engagement Con:** co-organized an industry-academia workshop on proprietary ranking-intervention results and feedback integration in platform systems.
- Apple** — Analytics and Machine Learning Intern Winter 2023
- Built** a graph neural network (Node2Vec, TensorFlow Keras) generating context-driven next-step recommendations via second-order random walks.
 - Integrated** a real-time Streamlit interface that updated predictions on adjustable parameters.
- The Lauder Institute, University of Pennsylvania** — Science and Technology Policy Intern Fall 2022
- Curated** 150+ think tanks and 100+ academic sources to the Global Go To Think Tank Index; analysis

featured by the World Economic Forum and the UN Center for Development.

Woebot Inc. — Machine Learning and AI Intern

Summer 2022

- **Spearheaded** topic modeling (DistilBERT, BERTopic, HDBSCAN, NLTK) over user messaging to ensure content relevance, producing a 23% improvement in user experience.
- **Retrained** classifier models to >90% precision and recall across content domains, and built software to automatically alert developers to emerging classifier categories.

Publications

Peer-Reviewed

Auditing Emergent Demographic Disparity in Misaligned Medical Language Models

2026

S. Pandey. Accepted to COLM 2026 Deploying AI in Healthcare Workshop.

Pitfalls in Evaluating Interpretability Agents

2026

T. Haklay, N. Prakash, **S. Pandey**, A. Torralba, A. Mueller, J. Andreas, T. Rott Shaham, Y. Belinkov. Accepted to BlackboxNLP 2026. arXiv:2603.20101

What We Know About Non-Engagement Signals in Content Ranking

2025

T. Cunningham, **S. Pandey**, L. Sigerson, J. Stray, J. Allen, B. Barrilleaux, R. Iyer, S. Milli, M. Kothari, B. Rezaei. New York Academy of Sciences, Special Annals 2025. arXiv:2402.06831

A StrongREJECT for Empty Jailbreaks

2024

A. Souly, Q. Lu, D. Bowen, T. Trinh, E. Hsieh, **S. Pandey**, P. Abbeel, J. Svegliato, S. Emmons, O. Watkins, S. Toyer. NeurIPS 2024; ICLR 2024 R2FM Workshop. arXiv:2402.10260

Forthcoming / In Progress

Evaluating for the Long Term: Learnings from Industry

2026

Leif Sigerson, Tom Cunningham, Winston Chou, **Sana Pandey**, Jonathan Stray, et. al. Whitepaper. arXiv:2608.08043v1.

Circuits Predict Cross-format Generalization in Arithmetic Reasoning

2025

A. de Varda, **S. Pandey**, P. Han. Submitted to ACL 2027.

Technical Skills

Languages & Frameworks: Python, SQL; PyTorch, JAX, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, pandas/NumPy.

Fine-tuning & Serving: LLM fine-tuning (PEFT/LoRA), distributed / multi-GPU training (Slurm), vLLM inference serving.

Safety, Evals & Interpretability: safety & jailbreak evaluations, red-teaming, benchmark design, evaluation harnesses (Inspect), mechanistic interpretability (TransformerLens, activation patching, sparse autoencoders), white-box evaluation.

Machine Learning: NLP (DistilBERT, BERTopic, HDBSCAN, NLTK), graph neural networks (Node2Vec), recommender & ranking systems, preference / human-feedback methods.

Awards & Fellowships

National Science Foundation Graduate Research Fellowship

2025

Harvard AI Safety Policy Fellowship

2026

Anthropic AI Research for Social Impact Fellowship (inaugural cohort, 8 selected)

2026

Graduation with Distinction, UC Berkeley

2024

USA Collegiate Fencing Nationals, Top 10

2024

UC Berkeley Dean's List

2021-2024

Talks & Service

Reviewer, BlackboxNLP

2026

Associate Editor, MIT Science Policy Review

2025-2026

Co-organizer, "Evaluating for the Long-Term" Workshop, MIT CoDE

2025

Panel Host, Cooperation and Coordination, CHAI Annual Workshop

2025

Panel Host, Societal Effects of AI, CHAI Annual Workshop

2024

Invited Speaker, UC Berkeley Haas School of Business

2022-2024

Media: CBS Evening News [🔗](#) (2023); UC Berkeley CDSS [🔗](#) (2024).

2023-2024